

# Where deep learning losses come from II

MLE, regression and classification: each loss is the negative log-score of a predictive distribution.

## THE QUESTION

Why is squared error the loss for regression, and cross-entropy the loss for classification?

## THE INTUITION

Assume a distribution for the target and write its negative log-likelihood. Gaussian noise around a line gives squared error; a Bernoulli label gives binary cross-entropy; a categorical label gives softmax cross-entropy.

## KEY TAKEAWAY

The distribution you assume decides the loss you get. MSE, BCE and CE are all  $-\log p(\text{observed target})$ .

distribution assumption  $\rightarrow$   $-\log p$  per example  $\rightarrow$  drop constants  $\rightarrow$  MSE · BCE · CE  $\rightarrow$  + prior  $\rightarrow$  MAP

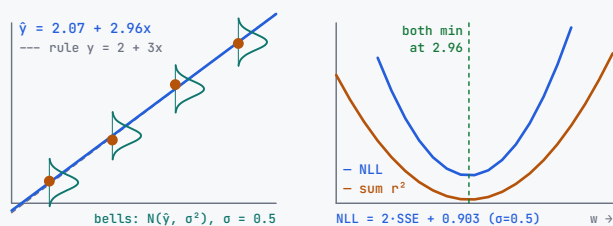
### 1 The shared derivation

Conditional independence gives  $p(\mathcal{D}|\theta) = \prod_i p_\theta(y_i|x_i)$ . Take logs and turn maximization into minimization:

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} - \sum_i \log p_\theta(y_i|x_i).$$

Define the one-example loss  $\ell_i(\theta) = -\log p_\theta(y_i|x_i)$ . The dataset objective is the sum or the mean of these contributions.

### 2 Gaussian likelihood gives squared error



Assume  $Y_i|x_i \sim \mathcal{N}(\mu_i, \sigma^2)$  with  $\mu_i = f_\theta(x_i)$ . Then

$$-\log p(y_i|x_i) = \frac{1}{2\sigma^2}(y_i - \mu_i)^2 + \log \sigma + C.$$

With  $\sigma$  fixed,  $\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} \sum_i (y_i - f_\theta(x_i))^2$ . On the four points above, the line (0, 1) scores sum  $r^2 = 56.58$ , the true rule (2, 3) scores 0.180, and the MLE (2.07, 2.96) scores 0.162.

MSE is Gaussian NLL up to a positive scale and additive constants. Learning  $\sigma$  requires keeping the  $\log \sigma$  term.

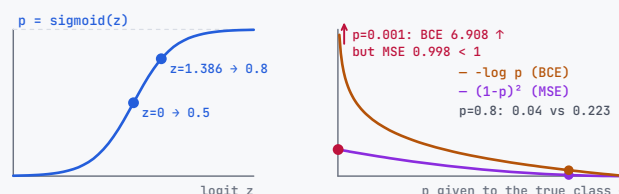
### 3 The residual model chooses the penalty

With residual  $r = y - \hat{y}$ :

|              | noise                     | NLL in $r$ | effect                |
|--------------|---------------------------|------------|-----------------------|
| Gaussian     | $r^2$                     |            | large errors dominate |
| Laplace      | $ r $                     |            | constant influence    |
| Student- $t$ | $\log(1 + r^2/(\nu s^2))$ |            | heavy-tail robustness |
| Uniform      | 0 in, $\infty$ out        |            | hard support          |

Constants and positive scale factors are omitted. Lecture 3 plots these shapes on data with outliers.

### 4 Bernoulli likelihood gives BCE



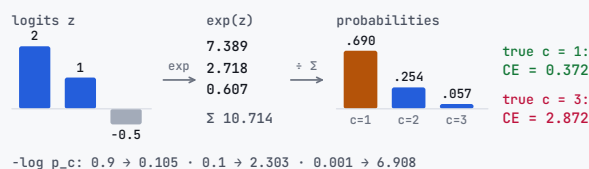
For  $y \in \{0, 1\}$  and predicted  $p = P(Y = 1|x)$ :  $p(y|x) = p^y(1-p)^{1-y}$ , so

$$\ell_{\text{BCE}} = -[y \log p + (1-y) \log(1-p)].$$

If  $y = 1$ ,  $p = 0.8$  gives  $\ell \approx 0.22$  while  $p = 0.2$  gives  $\ell \approx 1.61$ . MSE is possible but mismatched:  $(1-p)^2$  never exceeds 1, while  $-\log p$  is unbounded.

Output a logit  $z \in \mathbb{R}$  and use a fused BCE-with-logits operation for numerical stability.

### 5 Categorical likelihood gives cross-entropy



For mutually exclusive classes, logits  $z$  define  $p_k = \exp z_k / \sum_j \exp z_j$ . If the observed class is  $c$ ,

$$\ell_{\text{CE}} = -\log p_c = -z_c + \log \sum_j \exp z_j.$$

CE depends only on the probability given to the true class. Confident mistakes are expensive:  $-\log p_c \rightarrow \infty$  as  $p_c \rightarrow 0$ .

### 6 Match observation, output and loss

| target     | model output                    | NLL               |
|------------|---------------------------------|-------------------|
| real       | $\mu$ ; optionally $\sigma > 0$ | Gaussian / robust |
| binary     | one unconstrained logit         | Bernoulli BCE     |
| one of $K$ | $K$ unconstrained logits        | categorical CE    |
| count      | positive rate $\lambda$         | Poisson NLL       |

Before optimizing a loss, say which random variable and which distribution make it the right score.

# From MLE to MAP and regularization

Likelihood asks for data fit; a prior adds an explicit preference among plausible parameter settings.

## 7 Bayes combines data and prior

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

**likelihood** how the fixed data score each  $\theta$   
**prior** pre-data preference over  $\theta$   
**posterior** updated distribution after seeing the data  
**evidence** normalizer; independent of  $\theta$

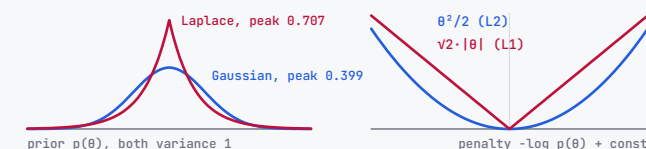
## 8 MAP is loss plus regularizer

$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta|\mathcal{D})$ . Drop the evidence and take negative logs:

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \left[ \underbrace{-\log p(\mathcal{D}|\theta)}_{\text{NLL}} + \underbrace{-\log p(\theta)}_{\text{prior penalty}} \right].$$

**MLE** uses only the likelihood. **MAP** returns one parameter vector, at the posterior mode.

## 9 Gaussian prior gives $L_2$



If  $\theta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ , then  $-\log p(\theta) = \frac{1}{2\tau^2} \|\theta\|_2^2 + C$ , so

$$\mathcal{L}_{\text{MAP}} = \text{NLL} + \lambda \|\theta\|_2^2, \quad \lambda = \frac{1}{2\tau^2}.$$

Smaller  $\tau$ : a narrower prior, larger  $\lambda$ , stronger shrinkage toward zero.

## 10 Laplace prior gives $L_1$

For independent zero-centred Laplace parameters with scale  $a$ ,  $-\log p(\theta) = \frac{1}{a} \|\theta\|_1 + C$ . The corner at zero can set coefficients **exactly** to zero:  $L_2$  shrinks smoothly,  $L_1$  can create sparsity.

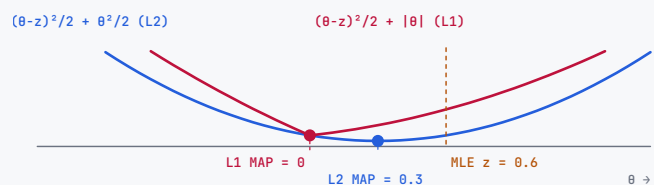
## 11 Scaling: keep the statistical meaning

The exact MAP objective uses summed NLL plus the full prior penalty:  $\sum_i \ell_i(\theta) + \lambda_{\text{sum}} R(\theta)$ . If code uses the mean NLL, divide the whole objective by  $n$ :

$$\frac{1}{n} \sum_i \ell_i(\theta) + \lambda_{\text{mean}} R(\theta), \quad \lambda_{\text{mean}} = \frac{\lambda_{\text{sum}}}{n}.$$

Changing the batch size should not silently change the intended regularization strength. Document whether a library returns a sum or a mean.

## 12 One-dimensional MAP shrinkage

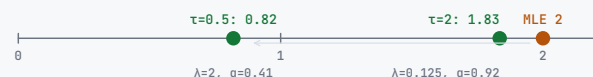


Approximate the likelihood by  $p(\mathcal{D}|\theta) \propto \exp(-(\theta - z)^2/(2s^2))$  with  $z = \hat{\theta}_{\text{MLE}}$ , and take  $\theta \sim \mathcal{N}(0, \tau^2)$ . Minimizing the two quadratics gives

$$\hat{\theta}_{\text{MAP}} = \underbrace{\frac{\tau^2}{\tau^2 + s^2}}_{\alpha \in (0,1)} z.$$

If  $z = 0.6$  and  $s = \tau$ , then  $\hat{\theta}_{\text{MAP}} = 0.3$ . With the  $L_1$  penalty  $|\theta|$  instead, the minimizer is exactly 0, because  $|z| < 1$ .

## 13 What stronger priors do



**More data** likelihood sharpens; MAP often approaches MLE  
**Narrower prior** posterior mode moves farther toward preferred values (above: MLE 2,  $s = 0.6$ )  
**Correlated prior** preferred directions become elliptical, not axis-aligned  
**Prior centre  $m$**  shrinkage is toward  $m$ , not automatically toward zero

## 14 Final audit

- ✓ Each loss is tied to an explicit likelihood.
- ✓ Constants are dropped only when they do not affect the optimized parameters.
- ✓ Logits and constrained distribution parameters are not confused.
- ✓ Prior family and scale are stated; sum-versus-mean scaling is explicit.

## 15 In code: weight decay is a Gaussian prior

```
nll = F.mse_loss(pred, y)      # mean NLL
pen = sum((w**2).sum()
           for w in model.parameters())
loss = nll + lam * pen          # MAP (lambda_mean)
# same gradient: SGD(..., weight_decay=2*lam)
```

Plain SGD's `weight_decay = 2λ` adds the gradient of  $\lambda \|\theta\|_2^2$ . AdamW's decoupled decay is not exactly this penalty.

**Likelihood gives the data loss · the prior gives the regularizer · their posterior mode gives MAP**

Page 1 numbers from the Lecture 2 storyboard data (four points from  $y = 2 + 3x$ ; logits (2, 1, -0.5)). Page 2 figures use the Lecture 3 storyboard's prior examples. Displayed values rounded. Slides, notebooks and summary videos: [nipunbatra.github.io/dl-2026](https://nipunbatra.github.io/dl-2026).